

HYDROGENATED NBR (HNBR) SYNTHESIS, DEGRADATION, AND PROPERTY-OPTIMIZATION PATENT LANDSCAPE

1. ABSTRACT

This report synthesizes a set of patents relevant to hydrogenated nitrile rubber (HNBR), with emphasis on how precursor NBR is hydrogenated, how HNBR molecular weight and residual unsaturation are controlled, and how downstream formulations are tuned for high-temperature performance. The target material is hydrogenated or partially hydrogenated nitrile-butadiene rubber, typically defined by reduced residual double bonds, improved heat/oil/ozone resistance, and processability suitable for seals, hoses, belts, and related elastomeric articles. Across the selected patents, two major synthesis themes emerge. First, direct hydrogenation of NBR is disclosed using homogeneous catalysts such as Wilkinson-type rhodium systems or ruthenium-based catalysts, often in solution and sometimes with co-catalysts or additives to manage selectivity and catalyst removal. Second, a more advanced route combines metathesis-based chain scission with hydrogenation, enabling lower molecular weight HNBR with narrower molecular weight distribution and improved processing behavior. These metathesis/hydrogenation processes use ruthenium or osmium carbene catalysts, sometimes with co-olefins, and in one case are accelerated by alkaline earth metal chloride additives. A third cluster of patents does not change the polymer backbone chemistry but instead improves HNBR compound performance through formulation: peroxide curing with unsaturated carboxylic acids/salts, microgels, non-reinforcing mineral fillers, basic silicates, and phosphine/phosphine oxide management. The landscape therefore spans precursor polymer design, catalytic hydrogenation, simultaneous metathesis-hydrogenation, and post-polymer compounding strategies. The strongest technical trends are toward lower Mooney viscosity, lower residual double bond content, narrower polydispersity, and improved compression set or high-temperature mechanical retention. Overall, the evidence supports HNBR synthesis routes that either preserve backbone integrity while hydrogenating selectively, or intentionally reduce molecular weight before/while hydrogenating to improve processability, followed by formulation strategies that preserve modulus and aging resistance in demanding automotive and industrial applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis; it discloses vulcanizable compositions based on hydrogenated nitrile rubber, peroxide, unsaturated carboxylic acid/salt, and specific microgels. The HNBR is described as a hydrogenated nitrile rubber precursor used in compounding, with the invention focused on improving high-temperature vulcanizate properties rather than polymer preparation. - Hydrogenated nitrile rubber content: 100 parts by weight based on 100 parts by weight of rubber — Claim 1 - Mooney viscosity (ML 1+4 @ 100°C): 10 to 120 MU — Claim 1 - Degree of hydrogenation of diene units: 95 to 100% — Claim 1 - Unsaturated carboxylic acids/salts: 1 to 100 parts by weight based on 100 parts by weight of rubber — Claim 1 - Peroxide: 0.2 to 8 parts by weight based on 100 parts by weight of rubber — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel amount: up to 60 parts by weight based on 100 parts by weight of rubber — Claim 2 - Microgel glass transition temperature: less than or equal to -50°C — Claim 3 - Mooney viscosity (ML 1+4 @ 100°C): 15 to 100 MU — Claim 4 - High-temperature storage condition: 175°C for 168 hours — Description - Hot-air storage condition: 150°C — Description

Relevant Experimental Evidence - Example 1: The patent's example section is not structurally parsed into discrete experimental runs, but the text emphasizes that microgels improve high-temperature mechanical properties of HNBR vulcanizates. - Example 1: The composition concept combines HNBR, unsaturated carboxylic acid/salt, peroxide, and microgels to retain hardness while improving modulus at elevated temperature. - Example 1: The disclosure highlights improved demoulding and reduced mould soiling in injection-moulding processes.

Technical Relevance This patent is highly relevant to HNBR application engineering because it defines a formulation window for peroxide-cured HNBR systems with microgels that improve compression set and high-temperature durability. It does not synthesize HNBR, but it directly informs how hydrogenated NBR is compounded for demanding service conditions.

Patent 2

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it discloses catalyst systems for metathesis reactions, including nitrile rubber metathesis, using a metathesis catalyst together with salt additives such as LiBr or CsBr. The relevance to HNBR lies in the precursor degradation step, where NBR can be chain-shortened by metathesis before hydrogenation to form lower-molecular-weight HNBR. - Salt(s): metathesis catalyst weight ratio 0.01:1 to 10000:1 — Claim 1 - Catalyst metal: osmium or ruthenium — Claim 1 - LiBr addition: 1 mg — Example 7 - CsBr addition: 1 mg — Example 8 - Grubbs II catalyst: 10 mg — Example 7 - Chlorobenzene: 0.3 ml — Example 7 - CDCl₃: 0.2 ml — Example 7 - Diethyl diallylmalonate addition: 0.15 ml — Example 7 - Room temperature — Example 7 - Time points: 0, 30, 60 min — Example 7 - Conversion with LiBr: 21.3% at 30 min; 57.7% at 60 min — Example 7 - Conversion without salt: 55.4% at 30 min; 100% at 60 min — Example 7 - Conversion with CsBr: 13.4% at 15 min; 25.3% at 30 min; 46.5% at 60 min; 71.9% at 90 min; 96.2% at 150 min — Example 8 - Conversion without salt: 16.5% at 15 min; 40.3% at 30 min; 68.9% at 60 min; 84.7% at 90 min; 100% at 150 min — Example 8

Relevant Experimental Evidence - Example 7: LiBr and CsBr were tested as salt additives in ring-closing metathesis, showing that salt addition changes reaction kinetics. - Example 8: CsBr was shown to alter conversion progression relative to the no-salt control over 150 minutes. - Example 7: The examples demonstrate the general principle that salt additives can modulate metathesis catalyst activity.

Technical Relevance This patent is relevant because it provides a catalyst-additive strategy that can be applied to metathesis-based NBR degradation prior to hydrogenation. Such control over metathesis activity is important for producing low-MW HNBR precursors with manageable reaction rates and selectivity.

Patent 3

- Patent Number: EP3196239A1
- Patent Title: Vulcanisable compounds on the basis of hydrogenated nitrile rubber, method for ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis; it discloses vulcanizable HNBR compositions containing a hydrogenated nitrile rubber, a non-reinforcing mineral filler, a basic silicate, and a peroxide-based crosslinking system. The technical contribution is formulation-based, targeting low compression set and high aging resistance at elevated temperature. - Residual double bonds (RDB): 10% or less — Claim - Residual double bonds (RDB): 7% or less — Claim - Residual double bonds (RDB): 1% or less — Claim - Non-reinforcing mineral filler: 50 to 100 parts by weight per 100 parts by weight of hydrogenated nitrile rubber — Description - Basic silicate: 5 to 50 parts by weight per 100 parts by weight of hydrogenated nitrile rubber — Description - Basic silicate pH in water (5 wt.%): greater than 7 — Description - Peroxide compound: 1 to 20 parts by weight per 100 parts by weight of hydrogenated nitrile rubber — Description - High-temperature compression set target: less than 45% after 175°C for 168 hours — Description - Preferred high-temperature compression set target: less than 40% after 175°C for 168 hours — Description - Hardness change target: 10% or less after 175°C for 168 hours — Description - Preferred hardness change target: 8% or less after 175°C for 168 hours — Description - Elongation-at-break change target: less than 35% after 175°C for 168 hours — Description - Preferred elongation-at-break change target: less than 30% after 175°C for 168 hours — Description - Preferred non-reinforcing mineral filler: 60 to 90 parts by weight — Description - Preferred basic silicate: 10 to 30 parts by weight — Description - Preferred peroxide compound: 2 to 10 parts by weight — Description

Relevant Experimental Evidence - Example 1: The patent establishes that combining a non-reinforcing mineral filler with a basic silicate in HNBR compounds improves high-temperature compression set and aging resistance. - Example 1: The disclosure frames the target as meeting demanding EN 549-type sealing requirements. - Example 1: The examples are formulation-focused rather than polymer-synthesis-focused.

Technical Relevance This patent is relevant to HNBR because it addresses the compounding of hydrogenated nitrile rubber for seals and similar parts requiring low compression set and high thermal aging resistance. It complements synthesis patents by defining a practical formulation window for high-performance HNBR vulcanizates.

Patent 4

- Patent Number: EP3219687B1
- Patent Title: Laminate and laminated glass
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it concerns laminates and laminated glass using a thermoplastic elastomer layer, including

hydrogenated products of block copolymers, rather than hydrogenated nitrile rubber. It is only tangentially related through the use of hydrogenated elastomeric materials in multilayer structures.

- Layer B viscosity average polymerization degree: 1000 — Example 6 - Layer B degree of acetalization: 70 mol% — Example 6 - Layer B plasticizer content: 30 parts by mass — Example 6 - Layer B shear storage modulus at 25°C: 0.70 MPa — Example 6 - Layer A styrene content: 12% by mass — Example 6 - Layer A peak temperature of tan δ : -22.6°C — Example 6 - Layer A peak height of tan δ : 1.92 — Example 6 - Laminate shear storage modulus at 25°C: 0.63 MPa — Example 6 - Laminate shear storage modulus at 50°C: 0.21 MPa — Example 6 - Maximum loss factor at 2,000 Hz, 1 to 50°C: 0.32 — Example 6 - Thickness ratio (layer A)/(total thickness of layer B): 1/3.75 — Example 6 - Breaking strength at 25°C: 0.23 kN — Example 6 - Lamination aptitude: C — Example 6 - Transmittance at 1,500 nm: 76 — Example 6 - Layer B viscosity average polymerization degree: 1700 — Comparative Example 11 - Layer B degree of acetalization: 70 mol% — Comparative Example 11 - Layer B plasticizer content: 15 parts by mass — Comparative Example 11 - Layer B plasticizer content: 60 parts by mass — Comparative Example 11 - Layer B shear storage modulus at 25°C: 82.0 MPa — Comparative Example 11 - Layer A styrene content: 12% by mass — Comparative Example 11 - Layer A peak temperature of tan δ : -22.6°C — Comparative Example 11 - Layer A peak temperature of tan δ : 0.7°C — Comparative Example 11 - Layer A peak temperature of tan δ : 35.2°C — Comparative Example 11 - Maximum loss factor at quaternary resonance frequency: 0.23, 0.36, 0.44, 0.037 — Comparative Example 11 - Breaking strength of laminated glass at 25°C: 0.15, 0.17, 0.15, 1.04 kN — Comparative Example 11

Relevant Experimental Evidence - Example 6: Demonstrates laminate acoustic and mechanical performance using a thermoplastic elastomer layer and polyvinyl acetal layers. - Comparative Example 11: Shows the effect of plasticizer content and layer properties on lamination aptitude and acoustic performance. - Example 12: Varies plasticizer content and layer thickness ratio to tune laminate modulus and transmittance.

Technical Relevance This patent is only indirectly relevant to HNBR because it uses hydrogenated elastomer concepts in a different application domain, not hydrogenated nitrile rubber synthesis. It is included because the manifest requires it, but it does not advance HNBR polymer preparation.

Patent 5

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of hydrogenated branched conjugated diene copolymers by first copolymerizing a branched conjugated diene compound with a vinyl compound, then hydrogenating the resulting copolymer. The hydrogenated copolymer is used in rubber compositions to improve processability and tire performance, but it is not an HNBR process. - 3-liter pressure resistant stainless steel vessel — Preparation Example 1 - Hexane: 2000 ml — Preparation Example 1 - Butadiene: 110 g — Preparation Example 1 - Styrene: 90 g — Preparation Example 1 - TMEDA: 0.22 mmol — Preparation Example 1 - n-Butyllithium: 35 mmol — Preparation Example 1 - Polymerization time: 5 hours — Preparation Example 1 - Reaction temperature: 50°C — Preparation Example 1 - Isopropanol/hexane quench: 1.15 ml of 1 M solution — Preparation Example 1 - Hydrogenation catalyst: 10 g of 10% palladium carbon — Preparation Example 2 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 2 - Hydrogenation temperature: 80°C — Preparation Example 2 - Degree of hydrogenation: 50% — Preparation Example 2 - Hexane: 500 ml — Preparation Example 3 - THF: 46 g — Preparation Example 3 - n-Butyllithium: 45 mmol — Preparation Example 3 - Isoprene: 150 g — Preparation Example 3 - Styrene: 125 g — Preparation Example 3 - Dropwise addition time: 2 hours — Preparation Example 3 - Isopropanol/hexane quench: 20 ml of 2 M solution — Preparation Example 3 - Hexane: 500 ml — Preparation Example 5 - THF: 46 g — Preparation Example 5 - n-Butyllithium: 40 mmol — Preparation Example 5 - Myrcene: 150 g — Preparation Example 5 - Styrene: 125 g — Preparation Example 5 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 5 - Hexane: 500 ml — Preparation Example 7 - THF: 46 g — Preparation Example 7 - n-Butyllithium: 40 mmol — Preparation Example 7 - Butadiene: 95 g — Preparation Example 7 - Myrcene: 55 g — Preparation Example 7 - Styrene: 125 g — Preparation Example 7 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 7 - Hydrogenation catalyst: 10 g of 10% palladium carbon — Preparation Example 12 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 12 - Hydrogenation temperature: 80°C — Preparation Example 12 - Degree of hydrogenation: 50% — Preparation Example 12

Relevant Experimental Evidence - Preparation Example 1: Demonstrates anionic solution copolymerization of butadiene/styrene followed by tire compounding and vulcanization. - Preparation Example 2: Shows hydrogenation of the copolymer using 10% Pd/C at 5.0 kgf/cm² and 80°C to give 50% hydrogenation. - Preparation Example 7: Demonstrates branched copolymer synthesis using butadiene, myrcene, and styrene. - Preparation Example 12: Repeats hydrogenation of a precursor copolymer under the same Pd/C, 5.0 kgf/cm², 80°C conditions. - Preparation Example 15: Shows synthesis of a farnesene/styrene copolymer precursor for subsequent hydrogenation.

Technical Relevance This patent is relevant only as a hydrogenated elastomer comparator, not as an HNBR disclosure, because it uses branched diene copolymers rather than nitrile rubber. It nevertheless illustrates solution polymerization and hydrogenation conditions that are conceptually adjacent to HNBR precursor modification.

Patent 6

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis-based degradation of nitrile rubber to produce low molecular weight hydrogenated nitrile rubber with narrower molecular weight distribution. The nitrile rubber is dissolved, contacted with a co-olefin such as ethylene or 1-hexene, and treated with a metathesis catalyst before or during hydrogenation to obtain low-MW HNBR. - Rubber: 200 g — Example 1 - MCB solvent: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber: 75 g — Example 3 - MCB solvent: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - 1-Hexene: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity of starting polymer: 73 — Table 2 - Mn of starting polymer: 85 kg/mol — Table 2 - Mw of starting polymer: 296 kg/mol — Table 2 - PDI of starting polymer: 3.50 — Table 2 - Mooney viscosity after Experiment 1: 43 — Table 2 - Mn after Experiment 1: 73 kg/mol — Table 2 - Mw after Experiment 1: 189 kg/mol — Table 2 - PDI after Experiment 1: 2.59 — Table 2 - Mooney viscosity after Experiment 2: 28 — Table 2 - Mn after Experiment 2: 60 kg/mol — Table 2 - Mw after Experiment 2: 136 kg/mol — Table 2 - PDI after Experiment 2: 2.27 — Table 2 - Mooney viscosity after Experiment 3: 3 — Table 2 - Mn after Experiment 3: 31 kg/mol — Table 2 - Mw after Experiment 3: 59 kg/mol — Table 2 - PDI after Experiment 3: 1.90 — Table 2 - Mooney viscosity after Experiment 4: 31 — Table 2 - Mn after Experiment 4: 55 kg/mol — Table 2 - Mw after Experiment 4: 111 kg/mol — Table 2 - PDI after Experiment 4: 2.02 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis of NBR cement at 6.9×10^5 Pa and 15 wt.-% solids was demonstrated. - Example 3: Lower solids loading (6 wt.-%) was used to further reduce molecular weight. - Example 4: 1-Hexene was used as the co-olefin instead of ethylene. - Table 2: The metathesis route reduced Mooney viscosity from 73 to as low as 3 and narrowed PDI to 1.90. - Table 2: The process produced HNBR with Mw down to 59 kg/mol and Mn down to 31 kg/mol.

Technical Relevance This patent is directly relevant to HNBR synthesis because it provides a route to low-molecular-weight HNBR via metathesis degradation followed by hydrogenation. It establishes practical catalyst loading, temperature, and co-olefin conditions that strongly influence processability and molecular-weight control.

Patent 7

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the same general metathesis-degradation and hydrogenation strategy for producing low molecular weight HNBR as the related US7772328B2 family, using nitrile rubber, a co-olefin, and metathesis catalysts. The process is aimed at lowering Mooney viscosity and narrowing molecular weight distribution before or during hydrogenation. - Rubber: 200 g — Example 1 - MCB solvent: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber: 75 g — Example 3 - MCB solvent: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - 1-Hexene: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity of starting polymer: 73 — Table 2 - Mn of starting polymer: 85 kg/mol — Table 2 - Mw of starting polymer: 296 kg/mol — Table 2 - PDI of starting polymer: 3.50 — Table 2 - Mooney viscosity after Experiment 1: 43 — Table 2 - Mn after Experiment 1: 73 kg/mol — Table 2 - Mw after Experiment 1: 189 kg/mol — Table 2 - PDI after Experiment 1: 2.59 — Table 2 - Mooney viscosity after Experiment 2: 28 — Table 2 - Mn after Experiment 2: 60 kg/mol — Table 2 - Mw after Experiment 2: 136 kg/mol — Table 2 - PDI after Experiment 2: 2.27 — Table 2 - Mooney viscosity after Experiment 3: 3 — Table 2 - Mn after Experiment 3: 31 kg/mol — Table 2 - Mw after Experiment 3: 59 kg/mol — Table 2 - PDI after Experiment 3: 1.90 — Table 2 - Mooney viscosity after Experiment 4: 31 — Table 2 - Mn after Experiment 4: 55 kg/mol — Table 2 - Mw after Experiment 4: 111 kg/mol — Table 2 - PDI after Experiment 4: 2.02 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis of NBR cement at 6.9×10^5 Pa and 15 wt.-% solids was demonstrated. - Example 3: Lower solids loading (6 wt.-%) was used to further reduce molecular weight. - Example 4: 1-Hexene was used as the co-olefin instead of ethylene. - Table 2: The metathesis route reduced Mooney viscosity from 73 to as low as 3 and narrowed PDI to 1.90. - Table 2: The process produced HNBR with Mw down to 59 kg/mol and Mn down to 31 kg/mol.

Technical Relevance This patent is directly relevant to HNBR synthesis because it provides a route to low-molecular-weight HNBR via metathesis degradation followed by hydrogenation. It establishes practical catalyst loading, temperature, and co-olefin conditions that strongly influence processability and molecular-weight control.

Patent 8

- Patent Number: US7737233B2

- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions, including nitrile rubber degradation, using ruthenium or osmium carbene catalysts. It is not itself an HNBR synthesis patent, but it provides the catalyst platform used in metathesis-based NBR chain scission prior to hydrogenation. - Catalyst metal: osmium or ruthenium — Description - HNBR degree of hydrogenation: 50 to 100% — Description - HNBR Mooney viscosity: 55 to 120 — Description - HNBR Mn: about 200000 to 700000 — Description - Residual double bond content: 1 to 18% — Description - Lowest workable NBR feedstock Mooney viscosity: about 30 Mooney units — Description - Hydrogenated nitrile rubber obtained from such feedstock: about 55 Mooney units — Description - Mooney viscosity of degraded NBR before hydrogenation: less than 30 Mooney units — Description - Mn of degraded NBR before hydrogenation: less than 70000 g/mol — Description - HNBR Mw after metathesis/hydrogenation: 30000 to 250000 — Description - HNBR Mooney viscosity after metathesis/hydrogenation: 3 to 50 — Description - PDI after metathesis/hydrogenation: less than 2.8 — Description

Relevant Experimental Evidence - Example section: The patent explains why conventional HNBR feedstocks are too viscous and why metathesis degradation is needed. - Example section: It contrasts Grubbs I and Grubbs II catalyst behavior in NBR metathesis. - Example section: It identifies the need for alternative catalysts to improve hydrogenation efficiency and metathesis activity.

Technical Relevance This patent is relevant because it frames the technical problem of HNBR processability and motivates metathesis-based molecular-weight reduction before hydrogenation. It provides the catalyst classes and target property ranges that underpin low-MW HNBR manufacture.

Patent 9

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber to produce HNBR in a single step. NBR is reacted in hydrogen, optionally with a co-olefin, using ruthenium- or osmium-based metathesis catalysts, enabling concurrent chain scission and saturation to lower molecular weight and residual unsaturation. - Catalyst metal: ruthenium or osmium — Claim 1 - Hydrogenated nitrile rubber Mw: 20,000 to 250,000 — Description - Hydrogenated nitrile rubber Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - Hydrogenated nitrile rubber MWD/PDI: less than 2.8 — Description - NBR feed Mooney viscosity: 24 to 70 — Description - Preferred NBR feed Mooney viscosity: 28 to 40 — Description - Hydrogenated nitrile rubber Mooney viscosity (ML 1+4 @ 100°C): 3 to 50 — Claim 1 - Hydrogenated nitrile rubber Mw: 30,000 to 250,000 — Claim 1 - Hydrogenated nitrile rubber PDI: less than 2.5 — Claim 1 - NBR feed Mooney viscosity before degradation: less than 30 Mooney units — Description - NBR feed Mn before degradation: less than 70,000 g/mol — Description - Hydrogenation catalyst example: Grubbs III — Example 1 - Hydrogenation catalyst example: Wilkinson's catalyst — Example 1 - Hydrogenation catalyst example: Grubbs II — Example 1 - Catalyst amount: 0.057 phr — Example 1 - Catalyst amount: 0.080 phr — Example 1 - Catalyst amount: 0.06 phr — Example 1 - Catalyst amount: 0.055 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Metal content: 91.3 ppm — Example 1 - Reaction time: 0 min, 60 min, 120 min, 180 min, 240 min — Example 1 - RDB at 0 min: 100% — Example 1 - RDB at 60 min: 2.0%, 0.7%, 15.4%, 2.9% — Example 1 - RDB at 120 min: 0.4%, 0.5%, 3.1%, 0.6% — Example 1 - RDB at 180 min: 0.3%, 0.3%, 1.3%, 0.3% — Example 1 - RDB at 240 min: 0.6% — Example 1 - Mw of HNBR 1: 191541 g/mol — Example 1 - Mn of HNBR 1: 68643 g/mol — Example 1 - PDI of HNBR 1: 2.79 — Example 1 - Mooney viscosity of HNBR 1: 38.2 — Example 1 - Mw of HNBR 2: 135772 g/mol — Example 1 - Mn of HNBR 2: 58259 g/mol — Example 1 - PDI of HNBR 2: 2.33 — Example 1 - Mooney viscosity of HNBR 2: 20.2 — Example 1 - Mw of HNBR 3: 195304 g/mol — Example 1 - Mn of HNBR 3: 64130 g/mol — Example 1 - PDI of HNBR 3: 3.05 — Example 1 - Mooney viscosity of HNBR 3: 69.3 — Example 1 - Mw of HNBR 4: 144212 g/mol — Example 1 - Mn of HNBR 4: 59851 g/mol — Example 1 - PDI of HNBR 4: 2.45 — Example 1 - Mooney viscosity of HNBR 4: 32.6 — Example 1

Relevant Experimental Evidence - Example 1: Compared Grubbs III, Wilkinson's, and Grubbs II catalysts for simultaneous hydrogenation/metathesis. - Example 1: Showed rapid RDB reduction from 100% to below 1% in selected catalyst systems within 180 to 240 minutes. - Example 1: Demonstrated that HNBR molecular weight and Mooney viscosity can be tuned by catalyst choice. - Example 1: Grubbs III was reported to hydrogenate faster in the first 120 minutes than Grubbs II or Wilkinson's catalyst.

Technical Relevance This is one of the most directly relevant patents because it defines a one-step route to HNBR by simultaneous metathesis and hydrogenation. It provides concrete catalyst classes and property targets for producing lower-MW, lower-RDB HNBR with improved processability.

Patent 10

- Patent Number: EP3090002B1

- Patent Title: Hydrogenated nitrile rubber containing phosphine or diphosphane oxide
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenated nitrile rubber containing phosphine or diphosphane oxide and a controlled halogen content, with emphasis on preparation and downstream vulcanizate performance. The synthesis context is hydrogenation of NBR using phosphine-containing catalyst systems, followed by management of residual phosphine/phosphine oxide species in the isolated HNBR. - Phosphines/diphosphines content: 0 to 1.0 wt.% based on HNBR — Claim 1 - Phosphine oxides/diphosphine oxides content: 0.25 to 6 wt.% based on HNBR — Claim 1 - Total halogen content: 25 to 10,000 ppm — Claim 1 - TPP use in hydrogenation: 5.5 wt.% — Example 3 - Reagent: C₂H₅Br — Example 3-1 - TPP in isolated HNBR: 0.75 wt.% — Example 3-1 - TPP=O in isolated HNBR: 1.41 wt.% — Example 3-1 - Reagent: benzyl bromide — Example 3-2 - TPP in isolated HNBR: not detectable — Example 3-2 - TPP=O in isolated HNBR: 0.94 wt.% — Example 3-2 - TPP in isolated HNBR: 1.33 wt.% — Example 3-3 - TPP=O in isolated HNBR: 2.17 wt.% — Example 3-3 - TPP use in hydrogenation: 10 phr — Prior-art discussion of DE 25 39 132 A - pH of aqueous THF solution target: >5 — Prior-art discussion of US-A-4,965,323 - pH of aqueous THF solution target: >5.5 — Prior-art discussion of US-A-4,965,323 - pH of aqueous THF solution target: >6 — Prior-art discussion of US-A-4,965,323 - Hydrogenation catalyst amount in US-A-4,503,196: 2.5 to 40 wt.% — Prior-art discussion

Relevant Experimental Evidence - Example 3-1: Ethyl bromide was used during hydrogenation, giving 0.75 wt.% TPP and 1.41 wt.% TPP=O in isolated HNBR. - Example 3-2: Benzyl bromide led to non-detectable TPP in isolated HNBR and 0.94 wt.% TPP=O. - Example 3-3: No reagent addition gave 1.33 wt.% TPP and 2.17 wt.% TPP=O in isolated HNBR. - Example 3: The examples show that reagent choice affects residual phosphine and phosphine oxide levels in HNBR.

Technical Relevance This patent is relevant because it addresses a practical issue in HNBR synthesis: residual phosphine/phosphine oxide and halogen contamination after hydrogenation. Such residues can strongly affect vulcanizate modulus and compression set, so the patent informs purification and catalyst-selection strategies.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US2473280B1 and the metathesis-degradation patents (US7772328B2, US7919563B2, US7737233B2, US8957161B2) all target lower molecular weight HNBR precursors, but EP2473280B1 does so in a single simultaneous hydrogenation/metathesis step, whereas US7772328B2 and US7919563B2 use metathesis degradation followed by hydrogenation.
- US2473280B1 reports HNBR Mw of 20,000 to 250,000 and Mooney viscosity of 1 to 50, while US7737233B2 describes post-metathesis HNBR typically at Mw 30,000 to 250,000 and Mooney 3 to 50; the latter is a broader catalyst/process disclosure rather than a single optimized example set.
- US7772328B2 and US7919563B2 show strong catalyst/co-olefin dependence: ethylene at 6.9×10^5 Pa and 1-hexene at 150 g are both used, with reactor temperature 80°C and catalyst loadings of 0.05 to 0.25 phr.
- EP2473280B1 uses hydrogenation catalysts such as Grubbs III, Grubbs II, and Wilkinson's catalyst with catalyst amounts around 0.055 to 0.080 phr and metal contents of 65.6 to 91.3 ppm, emphasizing catalyst choice as a control knob for RDB reduction.
- US9321910B2 and EP3196239A1 are formulation patents rather than synthesis patents: both start from hydrogenated nitrile rubber and focus on high-temperature vulcanizate performance, using microgels or filler/silicate combinations to improve compression set and aging resistance.
- EP3090002B1 differs from the other HNBR patents by focusing on residual phosphine/phosphine oxide and halogen content after hydrogenation; it highlights that catalyst residue management is important for final vulcanizate properties.
- US8957161B2 provides a salt-additive strategy for metathesis catalyst activation, which is mechanistically relevant to the metathesis-based HNBR route but is demonstrated on model metathesis substrates rather than HNBR itself.
- US7737233B2 and EP2473280B1 both rely on ruthenium/osmium carbene metathesis catalysts, but EP2473280B1 integrates hydrogenation into the same process, whereas US7737233B2 focuses on catalyst systems and process rationale.

4. CONCLUSION

The patent set shows that the most effective HNBR synthesis strategies are either selective hydrogenation of NBR with careful residue control or metathesis-assisted molecular-weight reduction followed by hydrogenation, including one-step simultaneous metathesis/hydrogenation. For practical HNBR manufacture, the evidence supports using NBR feedstocks with manageable Mooney viscosity, ruthenium- or osmium-based metathesis catalysts, and co-olefins such as ethylene or 1-hexene when chain scission is desired. For high-performance vulcanizates, peroxide curing combined with unsaturated carboxylic acids/salts, microgels, or non-reinforcing mineral filler/basic silicate packages improves compression set and thermal aging. A technically supported synthesis window is HNBR with Mw roughly 20,000 to 250,000, Mooney viscosity below about 50, and low residual double bond content, followed by formulation strategies that preserve modulus and durability at 130 to 175°C.

5. REFERENCES

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